

## Stable isotope ratios in atmospheric CH<sub>4</sub>: Implications for seasonal sources and sinks

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[1] We report  $\delta D$  and  $\delta^{13}C$  measurements of atmospheric CH<sub>4</sub> from air samples collected from two locations in the United States. They are the mid continental site Niwot Ridge, Colorado (40°N, 105°W), and a Pacific coastal site receiving strong westerlies, Montaña de Oro, California (35°N, 121°W). Data from multiyear approximately bimonthly sampling provide information relating seasonal cycling of CH<sub>4</sub> sources and sinks in background air, record long-term trends in CH<sub>4</sub> mixing and isotope ratio related to the atmospheric CH<sub>4</sub> loading, and may indicate regional CH<sub>4</sub> sources. At Niwot Ridge,  $\delta D$ -CH<sub>4</sub> averaged  $-93.1 \pm 3.0\text{‰}$  from 1999 to 2001, while  $\delta^{13}C$ -CH<sub>4</sub> averaged  $-47.22 \pm 0.13\text{‰}$  from 1995 to 2001 with distinct seasonal cycles in both isotope ratios. At Montaña de Oro, atmospheric CH<sub>4</sub> was observed to be more depleted in  $^{13}C$  and D: Measured  $\delta D$ -CH<sub>4</sub> averaged  $-97.3 \pm 3.7\text{‰}$  from 2000 to 2001, while  $\delta^{13}C$ -CH<sub>4</sub> averaged  $-47.26 \pm 0.17\text{‰}$  from 1996 to 2001, and seasonal cycles were larger than those observed at Niwot Ridge. Mixing ratios observed at Montaña de Oro were higher on average than at Niwot Ridge. At both sites,  $\delta^{13}C$ -CH<sub>4</sub> was found to correlate poorly with mixing ratio, an indication that varying CH<sub>4</sub> sources are partly responsible for the  $\delta^{13}C$ -CH<sub>4</sub> seasonal signal. In contrast, a strong anticorrelation exists between  $\delta D$ -CH<sub>4</sub> and mixing ratio, with maxima and minima approximately 6 months out of phase, indicating a sensitivity of  $\delta D$  to sink processes. The dual isotopic constraint to atmospheric CH<sub>4</sub> seasonality implies that these midlatitude sites are annually influenced by a  $^{13}C$ -enriched CH<sub>4</sub> source(s) seasonally increasing in late spring and a  $^{13}C$ -depleted CH<sub>4</sub> source(s) seasonally increasing in late summer or early fall.

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### 1. Introduction

[2] Methane (CH<sub>4</sub>), the most abundant hydrocarbon in the atmosphere, plays an important role in tropospheric and stratospheric chemistry [e.g., Cicerone and Oremland, 1988]. It reacts with hydroxyl radical (OH), acting as a control on the oxidizing power of the atmosphere and affecting many trace species such as carbon monoxide (CO), ozone (O<sub>3</sub>), formaldehyde, atmospheric hydrogen, and water vapor. It also reacts with Cl, where it sequesters Cl from taking part in O<sub>3</sub> destruction in the stratosphere. Methane is also an important greenhouse gas [Ramanathan et al., 1985; Dickinson and Cicerone, 1986] with a concentration known to be increasing on a global scale since preindustrial times [e.g., Blake and Rowland, 1988;

Dlugokencky et al., 1994, 1998; Etheridge et al., 1998]. Although this increase has been variable on a year to year basis in recent decades, it is persistent and the specific reasons for either increases or decreases in its growth rate are not fully understood [e.g., Dlugokencky et al., 2001, 2003]. It is widely believed that the increasing concentration of CH<sub>4</sub> and other greenhouse gases affect climate, and that if the present emission trends continue, global temperature will rise [Hansen et al., 2000; Ehhalt et al., 2001; Shindell et al., 2005].

[3] Stable isotope information provides a major constraint on CH<sub>4</sub> sources and sink processes. Stevens and Rust [1982] first proposed that a mass-weighted stable carbon isotopic balance between CH<sub>4</sub> sources and sink processes and the  $\delta^{13}C$ -CH<sub>4</sub> value in the atmosphere would help constrain the CH<sub>4</sub> budget. (Stable isotope measurement results are nearly always expressed in  $\delta$  notation as in the equation  $\delta^{13}C = [(^{13}C/^{12}C)_{\text{sam}} / (^{13}C/^{12}C)_{\text{std}} - 1] \times 1000$ , using carbon as an example. Here  $^{13}C/^{12}C_{\text{sam}}$  is the carbon isotope ratio in a sample and  $^{13}C/^{12}C_{\text{std}}$  is the carbon isotope ratio of V-PDB, an established standard.) They used a flux-weighted average value for the  $\delta^{13}C$  ratio of CH<sub>4</sub> entering the atmosphere and allowed for a slight  $^{13}C$  enrichment in the resulting atmo-

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spheric CH<sub>4</sub> isotopic value because of a kinetic isotope effect (KIE) in the reaction with OH, the principle CH<sub>4</sub> sink.

[4] Resulting calculations by *Stevens and Rust* [1982] helped apportion CH<sub>4</sub> sources according to whether they were relatively <sup>13</sup>C-enriched or <sup>13</sup>C-depleted with respect to atmospheric CH<sub>4</sub>. Since then, additional measurements of atmospheric δ<sup>13</sup>C-CH<sub>4</sub> have further constrained the CH<sub>4</sub> budget, provided information on seasonal cycles in CH<sub>4</sub> sources and sink processes, and helped understand changes in CH<sub>4</sub> mixing ratio [e.g., *Wahlen et al.*, 1989; *Quay et al.*, 1991, 1999; *Lowe et al.*, 1991, 1999; *Thom et al.*, 1993; *Brenninkmeijer et al.*, 1995; *Gupta et al.*, 1996; *Tyler et al.*, 1999; *Miller et al.*, 2002; *Morimoto et al.*, 2006]. In most cases these studies included multiyear regularly scheduled CH<sub>4</sub> sampling of well-mixed background air for δ<sup>13</sup>C-CH<sub>4</sub> at a fixed surface site or sites and compared the atmospheric data to available information on the δ<sup>13</sup>C of CH<sub>4</sub> sources and the KIEs of multiple sink processes. Only a few studies have reported substantial sets of measurements of δD in atmospheric CH<sub>4</sub> [*Quay et al.*, 1999; *Bergamaschi et al.*, 2000; *Rice et al.*, 2001, 2003]. Additional such atmospheric measurements would clearly be useful in studying changes in CH<sub>4</sub> and its growth rate, just as measurements of <sup>13</sup>C/<sup>12</sup>C ratios in CH<sub>4</sub> have proven to be.

[5] Here we present mixing and stable isotope ratio data of both δ<sup>13</sup>C and δD of CH<sub>4</sub> from long-term air sample collecting (approximately bimonthly) at two northern hemispheric sites, Niwot Ridge, Colorado (NWT) and Montaña de Oro, California (MDO). Measurement data from our air samples (December 1994 to November 2001 at NWT, January 1996 to December 2001 at MDO) show long-term trends and seasonal cycles of mixing, and δ<sup>13</sup>C and δD isotope ratios at both locations. Analysis of the data provides seasonal (winter/summer) temporal correlations, and offsets (deviations from 6-month lag or lead times) between isotope ratio values and CH<sub>4</sub> mixing ratios. These correlations illustrate the difference between the two isotopic tracers and, in particular, the strong influence of CH<sub>4</sub> sink reactions on atmospheric δD and the strong influence of CH<sub>4</sub> sources on atmospheric δ<sup>13</sup>C, thereby also illustrating the value of multiple stable isotope constraints on the CH<sub>4</sub> budget.

[6] We also present mixing ratios of carbon monoxide (CO) from the same air samples at both sites. In the Northern Hemisphere (NH), long-term trends in CO mixing ratio have gone from increasing (approximately 1980 to 1990) to decreasing (approximately 1991 to 2001) [*Wotowa et al.*, 2001; *Novelli et al.*, 2003]. The consensus reason for the decrease is that a decrease in NH fossil fuel emissions is responsible. Carbon monoxide has a strong influence on the oxidative capacity of the Earth's atmosphere. Hydroxyl radicals (OH), the most important oxidizing agent in the atmosphere, react with CO rather quickly, giving a CO lifetime of the order of a few months [*Levy*, 1971; *Logan et al.*, 1981; *Levine et al.*, 1985]. As for other trace gases, a knowledge of both the sources and sinks of tropospheric CO is vital to understanding spatial and temporal qualities of CO as well as for quantifying individual components of its budget such as OH concentrations. For our study of CH<sub>4</sub>, the CO mixing ratio data are of additional value because they help determine whether any of our air samples of well-

mixed background air have been influenced by local or regional sources.

## 2. Site Descriptions

[7] Niwot Ridge, Colorado (40°N, 105°W), is on the continental divide for North America. The air collection site is located about 4 km southeast of NWT at an altitude of approximately 3.15 km. It is situated on land owned by the Mountain Research Station of the University of Colorado and used as a Long-Term Ecological Research site by the National Science Foundation. Air samples are collected for us by the National Oceanic and Atmospheric Administration/Global Monitoring Division (NOAA/GMD) in Boulder, Colorado, through a collaborative agreement. Wind patterns at NWT are dominated by westerlies [*Barry*, 1973; *Haagenson*, 1979] although at times upslope flow from the south or southeast transports air from the Denver metropolitan area (located about 70 km to the southeast) to the site [*Hahn*, 1981; *Johnson and Toth*, 1982]. In our study all samples were collected before noon in order to increase the likelihood of sampling when the prevailing wind was from the west. Samples taken in this manner have the best chance of representing well-mixed background air from over the western United States without interference from regional contamination.

[8] Montaña de Oro State Park, a coastal site near San Luis Obispo (35°N, 121°W), is protected from the city of San Luis Obispo by low-lying foothills forming a ridge running parallel to the coast. The city of Morro Bay is northeast of the site, while the Diablo Canyon nuclear power plant is about 2 km to the southeast. According to the *California Energy Commission* [1985], which compiled measurements at two wind stations near the MDO site (Diablo Canyon, California, from years 1967 to 1981 and Santa Maria, California, from years 1948 to 1978), average annual wind speeds were about 16.1 and 12.7 km/hr, respectively, at the two stations and were predominately from the west and northwest. Additional data given for Santa Maria indicates that the months from March to June exhibited the highest winds, although only 20% or so above the mean wind speed, while late afternoon to early evening hours exhibited nearly twice the wind speed found for other times of the day. Accordingly, our air samples from MDO were collected in late afternoon or early evening during periods when prevailing winds were from the west or northwest.

## 3. Methods

### 3.1. Sample Collection Procedures

[9] Air samples were collected into either passivated aluminum high-pressure cylinders (NWT) and or into electropolished 32-L stainless steel canisters (MDO). The samples were pressurized using a compressor system which draws air through clean tubing and into the vessel after first passing through one or more drying tubes (magnesium perchlorate filled) located on the high-pressure side of the compressor. At NWT, the 793 liters (STP) of sample air was collected into the cylinders using a RIX model SA-3 compressor operated by our collaborators at NOAA/GMD. At MDO, canisters were filled by members of our research

group using a portable battery-operated piston pump (Model 415CDC30/12B, Thomas, Company, Sheboygan, Wisconsin). Details of the sampling procedure, including compressor cleaning, cylinder and canister conditioning, and drying tube preparation are given by Tyler *et al.* [1999].

### 3.2. Mixing Ratio Analytical Procedures

[10] At UC Irvine, CH<sub>4</sub> mixing ratios from canister and cylinder samples were measured using a Hewlett Packard 5880A gas chromatograph (GC). The analytical system included a 10-port valve, 2 mL sample loop, precolumn, separation column, and flame ionization detector. The separation column was 152 cm (3.18 mm o.d.) of molecular sieve 5A while the precolumn was 25 cm (3.18 mm o.d.) of the same material. Both columns were run isothermally at 100°C using nitrogen carrier gas with the precolumn back flushed contemporaneous to the latter part of the chromatographic separation period. For measuring CO mixing ratios we used a Shimadzu model 14A GC equipped with a 6-port valve, 5 mL sample loop, separation column, and reduction gas analyzer detector set at 275°C (model RGD2, Trace Analytical, Menlo Park, California). The separation column was 152 cm (3.18 mm o.d.) of molecular sieve 5A set at 80°C with a carrier gas of zero air. The separation column, detector, and carrier gas cleaning columns were conditioned between use as needed.

[11] Our mixing ratio working reference gases for CH<sub>4</sub> and CO analyses are based on NOAA/GMD reference scales for these gases and have been assigned values based on our primary reference gas. Our primary reference gas values of 1.903 ppm (CH<sub>4</sub>) and 134 ppb (CO) were directly intercompared with NOAA reference standards for CH<sub>4</sub> (CMDL83 scale [Lang *et al.*, 1990]) and for CO (CMDL scale WMO-88 [Novelli *et al.*, 1991, 2003]). Our level of precision for measurements are  $\pm 5$  to 10 ppb ( $1\sigma$ ,  $n = 5$ ) for CH<sub>4</sub> (approximately 0.25–0.50% uncertainty) and  $\pm 2$ –4 ppb ( $1\sigma$ ,  $n = 5$ ) for CO (approximately 1.0 to 2.0% uncertainty).

[12] We also intercompared our CH<sub>4</sub> mixing ratio results to those of NOAA/GMD for all our NWT air samples. The average difference between the values measured by NOAA and by our lab was  $3 \pm 7$  ppb ( $1\sigma$ ,  $n = 116$ ), with NOAA values higher. A significance test was made to establish the confidence limits on the difference between the two means. At the 95% confidence limit the 3 ppb difference was insignificant ( $\Delta x(\text{bar}) = 8$  ppb,  $n = 116$ ). No adjustment to NWT and MDO mixing ratio data presented here was made based on the small difference just noted. Note, however, that CH<sub>4</sub> mole fractions in the newest NOAA CH<sub>4</sub> standard gas scale (NOAA04) are a factor of  $1.0124 \pm 0.0007$  greater than those expressed in the CMDL83 scale [Dlugokencky *et al.*, 2005].

### 3.3. Isotope Ratio Analytical Procedures

[13] For  $\delta^{13}\text{C}$ -CH<sub>4</sub> analyses we used conventional dual inlet isotope ratio mass spectrometry (IRMS) methodology. In this method, air samples are processed using a combustion vacuum line, a necessary step to separate CH<sub>4</sub>, CO and CO<sub>2</sub> from other constituents of the air sample and to convert CH<sub>4</sub> to CO<sub>2</sub>. In brief, a series of cryotrap at  $-196^\circ\text{C}$  condense gases such as CO<sub>2</sub> and water vapor, leaving CO and CH<sub>4</sub> in the air stream. The CO is then converted to CO<sub>2</sub>

using I<sub>2</sub>O<sub>5</sub> on silica gel and trapped out at  $-196^\circ\text{C}$  before oxidizing CH<sub>4</sub> to CO<sub>2</sub> and H<sub>2</sub>O in a combustion furnace. Product CO<sub>2</sub> from CH<sub>4</sub> combustion can be separated from the H<sub>2</sub>O by-product using an ethanol/dry ice cryotrap. For a complete description of this procedure see Tyler *et al.* [1999] and references therein. Methane samples prepared as just described were then measured on our Finnigan MAT model 252 IRMS instrument. Precision of measurement on clean dry CO<sub>2</sub> gas standards was  $\pm 0.01\text{‰}$  ( $1\sigma$ ) for  $\delta^{13}\text{C}$  and  $\pm 0.05\text{‰}$  ( $1\sigma$ ) for  $\delta^{18}\text{O}$ . The reproducibility of  $\delta^{13}\text{C}$  measurements from approximately 200-L each of replicate air samples was  $\pm 0.05\text{‰}$  ( $1\sigma$ ) when all possible errors associated with differences in sample canisters, sample pumping, vacuum line processing, and isotope measurement were taken into account.

[14] For  $\delta\text{D}$ -CH<sub>4</sub> analyses we used our continuous flow (cf) GC-IRMS Thermo-Finnigan model Delta Plus XL instrument coupled to our custom-designed CH<sub>4</sub> gas preconcentrator. A pyrolysis oven converts CH<sub>4</sub> to H<sub>2</sub> after its separation from the air sample and before its detection by the mass spectrometer. For a comprehensive description of the CH<sub>4</sub> sample preparation and analyses, see Rice *et al.* [2001]. One procedural modification different than that described by Rice *et al.* was the decrease in required sample size from 120 to 63 mL of whole air, this serving to decrease sample processing times without loss of precision. Precision of measurement of  $\delta\text{D}$ -CH<sub>4</sub> was  $\pm 1.1\text{‰}$  ( $1\sigma$ ) based on pooled statistics from replicate measurements (3) of each sample. Long term precision and calibration drift was monitored through the routine measurement of a calibrated whole air sample. Combined, these uncertainties were estimated at  $\pm 1.3\text{‰}$  ( $1\sigma$ ).

### 3.4. Isotope Ratio Reference Gases

[15] Our working CO<sub>2</sub> isotope reference gas on the dual inlet IRMS instrument, assigned a value of  $-47.61\text{‰}$  versus V-PDB, was obtained from Dave Lowe of the National Institute for Water and Atmospheric Research (NIWA) in New Zealand. Another working CO<sub>2</sub> reference gas we purchased from Oztech Gas Company (Dallas, Texas) with assigned  $\delta^{13}\text{C}$  value of  $-39.78\text{‰}$  versus V-PDB was routinely compared to it. Both gases were been intercompared to two internationally recognized CO<sub>2</sub> standards, NBS-19 (CaCO<sub>3</sub>) and IAEA-CO-9 (BaCO<sub>3</sub>), which have established values of  $1.95\text{‰}$  and  $-47.12\text{‰}$  versus V-PDB-CO<sub>2</sub>, respectively [Stichler, 1995]. Versus our CO<sub>2</sub> reference gas at  $-47.61\text{‰}$ , clean dry CO<sub>2</sub> reference gases made in our laboratory from these carbonates had measured values of  $1.92\text{‰}$  for NBS-19 and  $-47.18\text{‰}$  for IAEA-CO-9.

[16] In an intercomparison of the long-term  $\delta^{13}\text{C}$ -CH<sub>4</sub> data set of D. Lowe and coworkers from samples taken at Baring Head, New Zealand [e.g., Lowe *et al.*, 1994, 1997, 1999], aliquots from 16 air samples were prepared and measured by both laboratories during 1998–1999 (8 samples from Montaña de Oro and 8 from Baring Head, New Zealand). The agreement for  $\delta^{13}\text{C}$ -CH<sub>4</sub> was  $+0.01 \pm 0.09\text{‰}$  ( $1\sigma$ ), with average values of samples measured at NIWA more  $^{13}\text{C}$ -enriched by  $+0.01\text{‰}$ .

[17] Results from another intercomparison program between our lab and NOAA/GMD on measurements of 10 NWT air samples collected from 2000–2001 indicate



agreement to within  $0.04 \pm 0.12\%$  ( $1\sigma$ ) between the two labs (with average values of samples measured at NOAA more  $^{13}\text{C}$ -enriched by  $+0.04\%$ ). Note, however, that measurements of  $\delta^{13}\text{C}$ -CH<sub>4</sub> made at NOAA are based on the scale used at UC Irvine [Miller *et al.*, 2002]. Therefore the close agreement in  $\delta^{13}\text{C}$  specifically validates sampling handling and processing for isotope analysis in the two labs rather than the absolute value of  $\delta^{13}\text{C}$  of CH<sub>4</sub>.

[18] On the cf-GC-IRMS, a working reference gas of ultrahigh purity (99.999%) H<sub>2</sub> from Oxygen Services Company (Costa Mesa, California) was assigned its value by comparison to a suite of three H<sub>2</sub> gas cylinders purchased from Oztech Gas Company. These have assigned  $\delta\text{D}$  values of  $-108.0$ ,  $-165.3$ , and  $-306.8\%$  versus VSMOW. Our cf-GC-IRMS H<sub>2</sub> working reference gas was assigned a  $\delta\text{D}$  value of  $-169.4 \pm 0.5\%$  ( $1\sigma$ ) versus VSMOW.

#### 4. Analyses of Trends

[19] Our NWT CH<sub>4</sub> and CO mixing ratio and  $\delta^{13}\text{C}$ -CH<sub>4</sub> data sets encompass the period from 29 December 1994 to 23 November 2001. Measurements of  $\delta\text{D}$ -CH<sub>4</sub> at NWT began on 28 August 1998 and are presented from that date through 23 November 2001. Air sampling at MDO began in 23 January 1996 with CH<sub>4</sub> and CO mixing ratio and  $\delta^{13}\text{C}$ -CH<sub>4</sub> isotope ratio data presented from inception to 1 December 2001. Measurements of  $\delta\text{D}$ -CH<sub>4</sub> at MDO are available only from 25 February 2000 to 1 December 2001. We have archived these data, to make them available for modeling and advanced calculations by other atmospheric researchers, on the internet Web site for the Carbon Dioxide Information Analysis Center at <http://cdiac.ornl.gov/home.html>. Our CH<sub>4</sub> and CO mixing ratio data in years 2000–2001 are also archived on the internet Web site for the MOPITT instrument/Terra Satellite program at <http://www.eos.ucar.edu/mopitt/validation/index.html>.

##### 4.1. Long-Term Trends in CH<sub>4</sub> and CO Mixing and CH<sub>4</sub> Isotope Ratio Data

[20] Figures 1a, 1b, 1c, and 1d show the CH<sub>4</sub> mixing ratio,  $\delta^{13}\text{C}$ -CH<sub>4</sub>,  $\delta\text{D}$ -CH<sub>4</sub>, and CO mixing ratio, respectively, at NWT for the time periods given previously (e.g., from 29 December 1994 to 23 November 2001 for mixing ratio). Similarly, Figures 2a, 2b, 2c, and 2d show the CH<sub>4</sub> mixing ratio,  $\delta^{13}\text{C}$ -CH<sub>4</sub>,  $\delta\text{D}$ -CH<sub>4</sub>, and CO mixing ratio, respectively, at MDO. Curve fits to mixing ratios,  $\delta^{13}\text{C}$ -CH<sub>4</sub> and  $\delta\text{D}$ -CH<sub>4</sub> are unweighted least squares fits to a quadratic polynomial (which represents the long-term trend) and additional harmonic terms (which represent the average seasonal cycle) expressed as equation (1) [e.g., Steele *et al.*, 1992].

$$y = a + bt + ct^2 + d \sin(2\pi t/365) + e \cos(2\pi t/365) + f \sin(4\pi t/365) + g \cos(4\pi t/365) \quad (1)$$

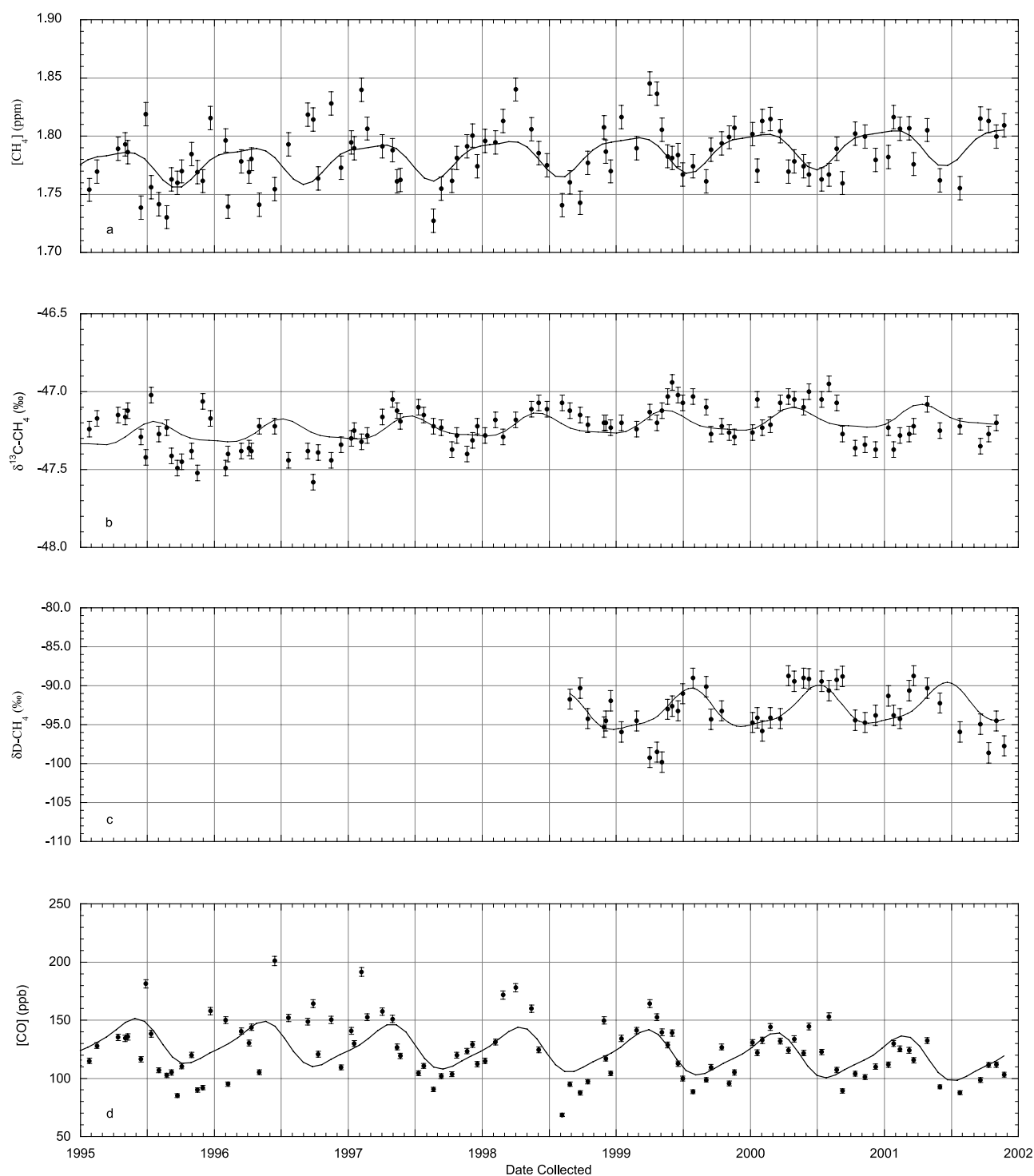
[21] Average mixing ratios at NWT and MDO over the entire sampling periods were  $1.785 \pm 0.029$  ppm ( $1\sigma$ ) and  $1.811 \pm 0.030$  ppm ( $1\sigma$ ), respectively. Mean growth rates of CH<sub>4</sub> for the period 1996–1999 (period with positive growth) were  $5.0 \pm 1.8$  ppb/yr ( $1\sigma$ ) at NWT and  $5.3 \pm 3.7$  ppb/yr ( $1\sigma$ ) at MDO, in good agreement with the growth rate of  $5.4$  ppb/yr ( $1\sigma$ ) reported for the  $30^\circ$ – $90^\circ\text{N}$

latitudinal zone by NOAA/GMD for the years 1996–1999 [Dlugokencky *et al.*, 2001]. Annual CH<sub>4</sub> growth rates at each site were determined by difference in mean values in successive years. At NWT mean growth rates were 10, 1, 4, 10,  $-11$ , and 11 ppb for 1995–1996 through 2000–2001, respectively. At MDO mean annual CH<sub>4</sub> growth rates were 1,  $-2$ , 17,  $-3$ , and  $-6$  ppb for 1996–1997 through 2000–2001, respectively. While both NWT and MDO showed positive change in CH<sub>4</sub> mixing ratio over the entire duration of our measurement record, data in Figures 1a and 2a show a slowing of growth during 2000 at both sites and continuing through the year 2001. The slowing of growth beginning in 2000 was in keeping with results reported by Simpson *et al.* [2002] whose data indicate globally averaged negative growth in CH<sub>4</sub> mixing ratio during 2000–2001, unprecedented in modern (post industrial revolution) time series of CH<sub>4</sub> measurements.

[22] Because of large variability in mixing ratio as a result of seasonal and irregular effects in these data relative to the number of measurements, observed trends in CH<sub>4</sub> growth rate were not generally large enough to be statistically significant at high levels of confidence (e.g.,  $>20$  ppb at 95% confidence interval). However, qualitatively both NWT and MDO showed a large increase in CH<sub>4</sub> growth for the interval 1998–1999 compared to surrounding yearly growth rate values, similar to that reported by Morimoto *et al.* [2006]. Our record of growth rate fluctuations may reflect the influence of the large ENSO occurrence during 1997–1998 and its reported effect on biomass burning and enhanced CH<sub>4</sub> release.

[23] The averages for measured values of  $\delta^{13}\text{C}$ -CH<sub>4</sub> at NWT and MDO were  $-47.22 \pm 0.13\%$  ( $1\sigma$ ) and  $-47.26 \pm 0.17\%$  ( $1\sigma$ ), respectively. At NWT, the  $\delta^{13}\text{C}$ -CH<sub>4</sub> trend averaged  $+0.03 \pm 0.01\%$ /year ( $1\sigma$ ) over the entire time period. However, in comparing annually averaged values for each year, a decrease from 1995 to 1996 was following by an increase from 1996 to 1997 after which changes were relatively small. Differences between successive annual mean values of  $\delta^{13}\text{C}$  at NWT were  $-0.13$ ,  $0.15$ ,  $0.06$ ,  $0.03$ ,  $-0.01$ , and  $0.10\%$  from 1995–1996 through 2000–2001, respectively. At MDO, the trend over the entire data record averaged  $+0.05 \pm 0.01\%$ /year ( $1\sigma$ ). Differences between successive annual mean  $\delta^{13}\text{C}$  values at MDO were  $0.06$ ,  $0.18$ ,  $0.05$ ,  $0.05$ , and  $0.01$  for 1996–1997 through 2000–2001, respectively. The overall trend of increasing  $^{13}\text{C}$ -enrichment in atmospheric CH<sub>4</sub> during this period is generally thought to be the result of an increased importance of  $^{13}\text{C}$ -enriched sources such as fossil fuels and biomass burning rather than the result of increases in any of the atmospheric chemical sinks, although some lag in the forcing of such sinks may change  $\delta^{13}\text{C}$ -CH<sub>4</sub> without a contemporaneous change in the actual sink [Tans, 1997]. Similar to changes in CH<sub>4</sub> mixing ratio, interannual fluctuations in  $\delta^{13}\text{C}$  were obscured by seasonal and irregular effects and mean differences were rarely significant at high levels of confidence, (e.g.,  $>0.09\%$  at 95% confidence interval).

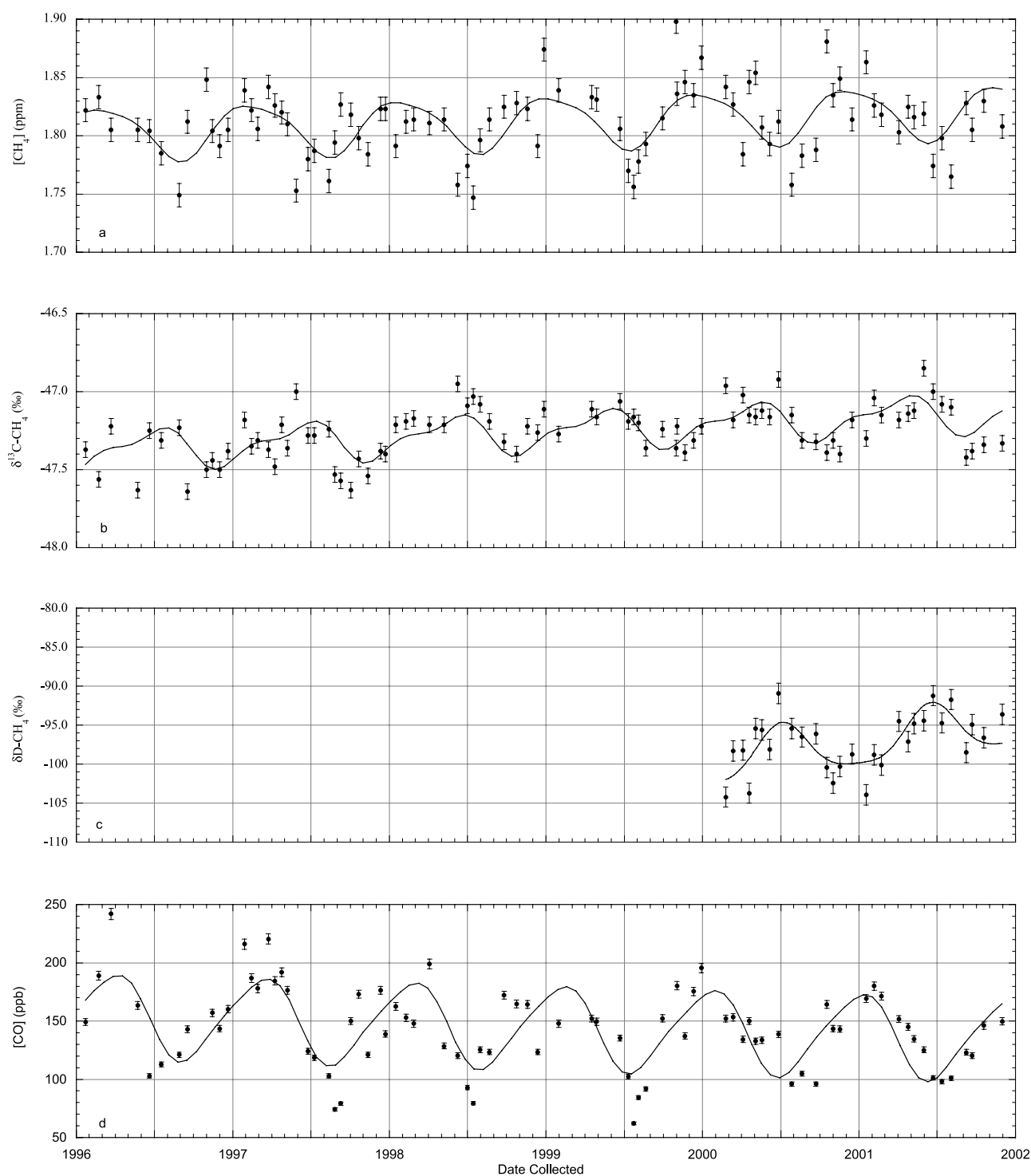
[24] Qualitatively the annual differences in  $\delta^{13}\text{C}$  do not appear to track changes in the CH<sub>4</sub> growth rate. Unlike our observation of CH<sub>4</sub> growth for the interval 1998–1999 relative to surrounding years, the annual change in  $\delta^{13}\text{C}$ -CH<sub>4</sub> at both NWT and MDO was not noticeably different



**Figure 1.** (a) CH<sub>4</sub> mixing ratio, (b)  $\delta^{13}\text{C}-\text{CH}_4$ , (c)  $\delta\text{D}-\text{CH}_4$ , and (d) CO mixing ratio at Niwot Ridge, Colorado, from 29 December 1994 to 23 November 2001. (The  $\delta\text{D}-\text{CH}_4$  data encompass 28 August 1998 to 23 November 2001.) Error bars are one standard deviation from the average of multiple samples. Curves fit to data are unweighted least squares fits (2-harmonic) of equation (1).

from 1998–1999 compared to other year intervals. As suggested by Morimoto *et al.* [2006], perhaps an anomalous CH<sub>4</sub> release from both <sup>13</sup>C-enriched biomass and fossil fuel burning as well as <sup>13</sup>C-depleted CH<sub>4</sub> from wetland emissions could have occurred.

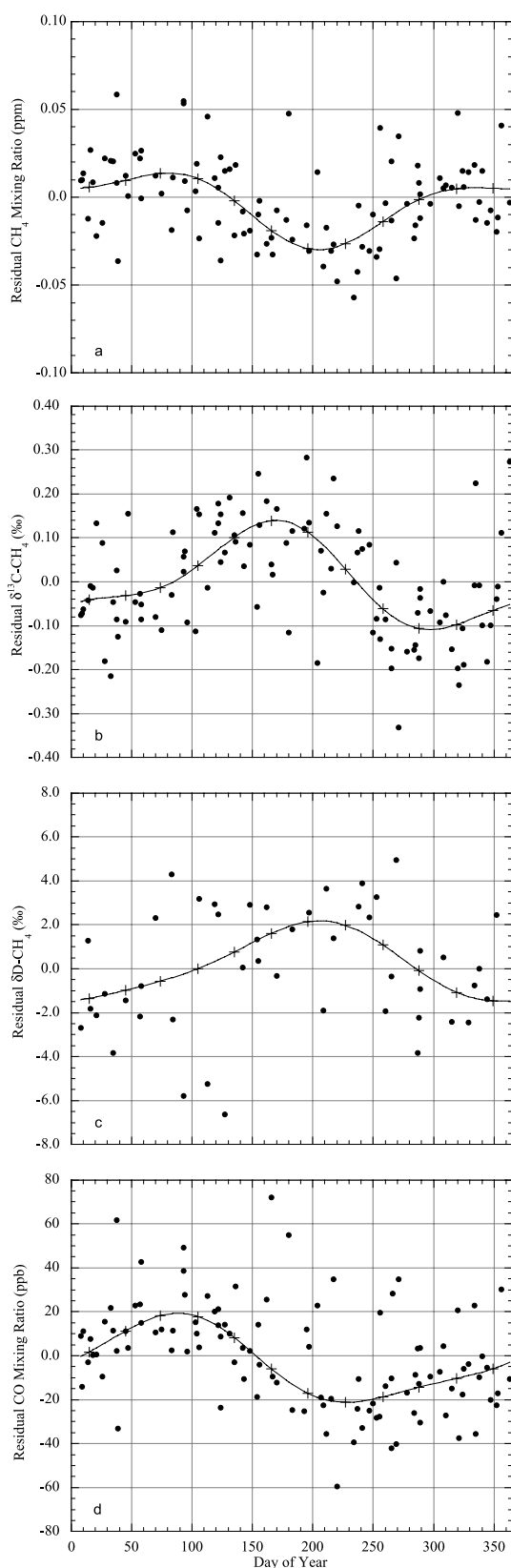
[25] Figures 1c and 2c plot the  $\delta\text{D}-\text{CH}_4$  data versus time. The three year record at NWT indicates a trend for  $\delta\text{D}$  between 28 August 1998 and 23 November 2001 of  $+1.4 \pm 3.9\text{‰/year}$  ( $1\sigma$ ). The average value of  $\delta\text{D}$  value at NWT over this period of time was  $-93.1 \pm 3.0\text{‰}$  ( $1\sigma$ ). For three dates in 1999 (2 April, 22 April, and 6 May), the  $\delta\text{D}$  values



**Figure 2.** (a) CH<sub>4</sub> mixing ratio, (b)  $\delta^{13}\text{C}\text{-CH}_4$ , (c)  $\delta\text{D-CH}_4$ , and (d) CO mixing ratio at Montaña de Oro, California, from 23 January 1996 to 1 December 2001. (The  $\delta\text{D-CH}_4$  data encompass 25 February 2000 to 1 December 2001.) Error bars are one standard deviation from the average of multiple samples. Curves fit to data are unweighted least squares fits (2-harmonic) of equation (1).

deviated considerably from the mean seasonal cycle. The first two of these dates were also relatively high in CH<sub>4</sub> mixing ratio as were all three dates for CO mixing ratio compared to the seasonal curve fit. However,  $^{13}\text{C}$  values were tightly coupled between observed and curve fit  $\delta^{13}\text{C}\text{-CH}_4$ . On the basis of an analysis of 5-day back trajectories

for all sampling dates at NWT, no obvious explanation exists for the extreme behavior in  $\delta\text{D-CH}_4$  on these three dates of collection (calculated using the HYSPLIT model of the NOAA Air Resources Lab, available at <http://www.arl.noaa.gov/ready/hysplit4.html>). The MDO  $\delta\text{D-CH}_4$  record covers less than two full years. The average value of



**Figure 3.** (a) Detrended CH<sub>4</sub> mixing ratio, (b) δ<sup>13</sup>C-CH<sub>4</sub>, (c) δD-CH<sub>4</sub>, and (d) CO mixing ratio at Niwot Ridge, Colorado. Curves fit to data are unweighted least squares Fourier fits with a fixed 1-year period and include the fundamental and first harmonic.

δD-CH<sub>4</sub> at MDO was  $-97.2 \pm 3.6\text{‰}$  ( $1\sigma$ ) from measurements spanning 25 February 2000 to 1 December 2001. Over this period, the observed trend was  $+2.6 \pm 2.0\text{‰/year}$  ( $1\sigma$ ). However, because the seasonal cycle is considerable at this site (see below) and the data spans less than two complete years, the trend may be biased by having relatively less winter (D depleted) data points in the time series. Clearly more data will be required to establish long-term δD-CH<sub>4</sub> trend at NWT and MDO.

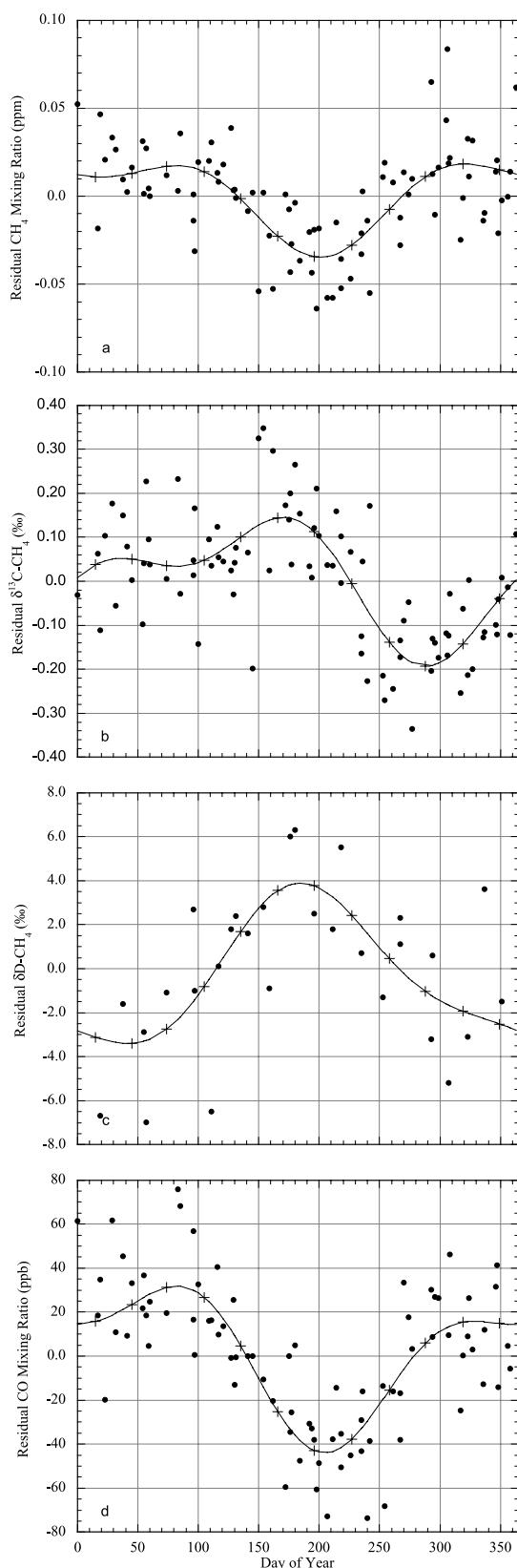
[26] Figures 1d and 2d plot the CO mixing ratio data versus time. Mean CO mixing ratio values were  $124 \pm 24$  ppb ( $1\sigma$ ) at NWT and  $146 \pm 35$  ppb ( $1\sigma$ ) at MDO. Over the same time span as our measurements at NWT, the NOAA average CO concentration calculated from monthly means was 122 ppb. Our data indicate that CO mixing ratio was decreasing at an average rate of  $-1 \pm 1$  ppb/yr ( $1\sigma$ ) at NWT (from 1995 through 2001) and an average rate of  $-4 \pm 2$  ppb/yr ( $1\sigma$ ) at MDO (from 1996 through 2001). At NWT an increase from 1995 to 1996 was followed by a sharp decrease over the next succeeding year and then very little yearly change until a sharp decrease from 2000 to 2001. However, at MDO the rate of decline slowed down each year. The mean annual CO mixing ratio differences at NWT were 19, -12, -4, -1, 1, and -11 ppb for successive years from 1995–1996 through 2000–2001, respectively, while at MDO they were 1, -14, -4, -2, and 3 from 1996–1997 through 2000–2001, respectively. Differences in observed mean annual growths in CO at each site were, in general, not significant at high levels of confidence (e.g., >16 ppb at 95% confidence interval).

#### 4.2. Seasonal Behavior in CH<sub>4</sub> and CO Mixing and CH<sub>4</sub> Isotope Ratio Data

[27] Data of CH<sub>4</sub> mixing ratio, δ<sup>13</sup>C-CH<sub>4</sub>, and δD-CH<sub>4</sub> were analyzed for seasonal behavior. Long-term trends were first removed by fitting each time series to a polynomial curve and detrending mixing and isotope ratio data with averages subtracted accordingly. The residuals were then plotted versus date of collection over a calendar year. Both Fourier series and loess [Cleveland *et al.*, 1990] curve fits provided very similar depiction of annual cycles. We choose to display the Fourier fits here since that method will be more familiar to most readers. The Fourier series have a fixed period of one year and include the fundamental and first harmonic. Robust statistical methods were used so as to avoid the strong influence of outliers. Inclusion of successive terms in the Fourier series yielded lower statistical significance (i.e.,  $p$  values > 0.1) accompanied with no considerable improvement in the qualitative description of seasonal behavior. Note that our goal here was to evaluate repeatable seasonal behavior at our two specific northern hemispheric sites.

[28] Figures 3a, 3b, 3c, and 3d show over 1 year residuals for the CH<sub>4</sub> mixing ratio, δ<sup>13</sup>C-CH<sub>4</sub>, δD-CH<sub>4</sub>, and CO mixing ratio, respectively, at MDO. In a similar way, Figures 4a, 4b, 4c, and 4d show residuals for the CH<sub>4</sub> mixing ratio, δ<sup>13</sup>C-CH<sub>4</sub>, δD-CH<sub>4</sub>, and CO mixing ratio, respectively, at NWT. Mid monthly values, interpolated from the fits, are plus symbols for January through December. Seasonal cycles in mixing ratio, δ<sup>13</sup>C, and δD were significantly larger at MDO than at NWT. Annual peak-to-peak amplitude in





**Figure 4.** (a) Detrended CH<sub>4</sub> mixing ratio, (b)  $\delta^{13}\text{C}\text{-CH}_4$ , (c)  $\delta\text{D}\text{-CH}_4$ , and (d) CO mixing ratio at Montaña de Oro, California. Curves fit to data are unweighted least squares Fourier fits with a fixed 1-year period and include the fundamental and first harmonic.

mixing ratio was 49 ppb at NWT and 62 ppb at MDO. Average annual isotope seasonal peak-to-peak amplitudes at the two sites were calculated to be 0.23‰ and 0.38‰, respectively, for  $\delta^{13}\text{C}\text{-CH}_4$  and 5.2‰ and 10.3‰, respectively, for  $\delta\text{D}\text{-CH}_4$ . Peak-to-peak amplitudes determined from data point extremes rather than from calculated fits can be about 30 to 50% larger for individual years.

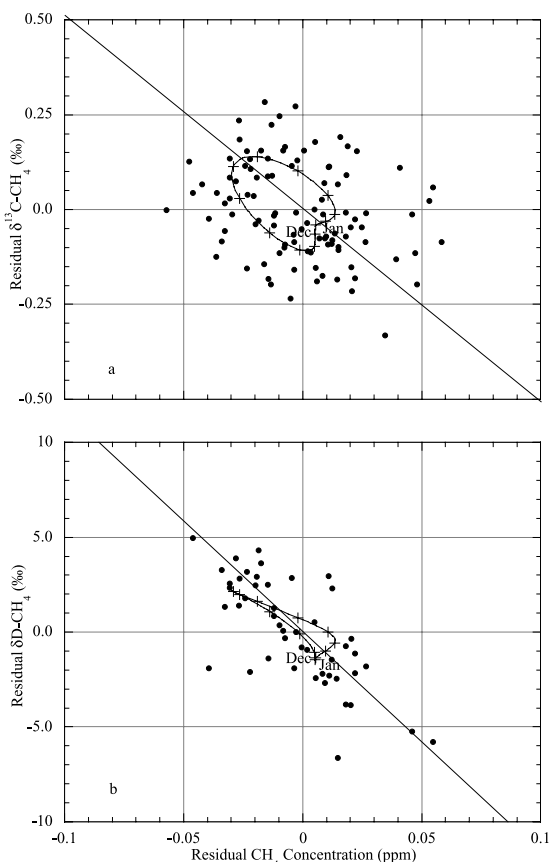
[29] Despite differences in both mean values and seasonal amplitude, seasonal phases were observed to be consistent between the two sampling sites, an indication of similar forcings. The CH<sub>4</sub> mixing ratio (Figures 3a and 4a) is fairly constant and near its maximum between late fall (October–November) through early spring (March–April). A sharp minimum in mixing ratio occurs regularly in middle to late summer (July–August). Seasonal behavior in  $\delta^{13}\text{C}\text{-CH}_4$  (Figures 3b and 4b) indicates that CH<sub>4</sub> has the most  $^{13}\text{C}$ -enriched CH<sub>4</sub> in early to middle summer (May–July) and fairly distinct  $^{13}\text{C}$ -depleted CH<sub>4</sub> in mid fall (October). Winter and early spring  $\delta^{13}\text{C}\text{-CH}_4$  fall in between summer and fall values. Fourier fits to both mixing ratio and  $\delta^{13}\text{C}$  at NWT and MDO reveal significant contributions from the fundamental cosine and sine terms. Of the first harmonic, only the cosine term makes a considerable contribution as it is symmetric about midyear (whereas the sine harmonic is antisymmetric about midyear). Correlations between mixing ratio and  $\delta^{13}\text{C}$  are typically weak (see discussion below). In contrast, Figures 3c and 4c both show that  $\delta\text{D}\text{-CH}_4$  behaves in antiphase with mixing ratio. Methane enriched in D is observed during the mixing ratio minimum in summer (June–August) while CH<sub>4</sub> depleted in D was observed during the late fall and winter when mixing ratio was at its maximum. The relationship between mixing ratio and  $\delta\text{D}$  values is discussed in detail below.

[30] The seasonal cycle in CO mixing ratio is larger at MDO (76 ppb, Figure 4d) than it is at NWT (40 ppb, Figure 3d), a result similar to that from observations of CH<sub>4</sub> at the two sites. However, seasonal CO phases were again observed to be similar between the sampling stations, as they were for CH<sub>4</sub>. At NWT a maximum in March and a minimum in August and September was observed for the CO mixing ratio. At MDO a broad maximum between November and May (with a slight dip in January) and a minimum in July and August was exhibited. These CO inflection points can be directly compared to the ones shown in Figures 3a and 4a for CH<sub>4</sub>. The CO maxima and minima features closely track those of CH<sub>4</sub> with the exception of CO at NWT having no dip in the maxima.

#### 4.3. Comparison With Other CH<sub>4</sub> and CO Data Sets

[31] Our data sets of  $\delta^{13}\text{C}$  and  $\delta\text{D}$  of CH<sub>4</sub> from NWT and MDO can be compared with respect to general features with other long-term CH<sub>4</sub> isotope data sets. Data sets compiled from southern hemispheric sites in the middle and high latitudes have consistently shown qualitatively opposite features of seasonality to be exhibited contemporaneously to our northern hemispheric sites (e.g., data from Baring Head, New Zealand, and South Pole reported on by Lowe *et al.* [1991, 1994, 1997]). However, average values of  $\delta^{13}\text{C}$  of CH<sub>4</sub> reported by Lowe *et al.* were always more positive (by 0.1 to 0.5‰ depending on time of year) in comparison to our Northern Hemispheric values over the same years. This is because the N/S difference in  $\delta^{13}\text{C}$  is driven particularly





**Figure 5.** Phase ellipses for CH<sub>4</sub> at Niwot Ridge, Colorado, made by plotting (a) residual  $\delta^{13}\text{C}$ -CH<sub>4</sub> versus residual CH<sub>4</sub> mixing ratio and (b) residual  $\delta\text{D}$ -CH<sub>4</sub> versus residual CH<sub>4</sub> mixing ratio. The slopes of the lines result in kinetic isotope effects for atmospheric loss processes of  $1.0090 \pm 0.003$  and  $1.30 \pm 0.03$  for  $k_{12}/k_{13}$  and  $k_{\text{H}}/k_{\text{D}}$ , respectively.

by the much larger proportion of emissions of CH<sub>4</sub> sources in the NH as compared to the Southern Hemisphere (SH).

[32] Quay *et al.* [1999] reported on mixing ratio and  $\delta^{13}\text{C}$  and  $\delta\text{D}$  of CH<sub>4</sub> with time series records at 6 sites between 1988 and 1995. The trend in  $\delta^{13}\text{C}$  reported over this period was  $0.02\text{‰/yr}$ , similar to the trend of increasing  $^{13}\text{C}$ -enrichment in atmospheric CH<sub>4</sub> found at our sites. The amplitude of the seasonal cycle in  $\delta^{13}\text{C}$  ranged from about  $0.4\text{‰}$  at  $71^\circ\text{N}$  to  $0.1\text{‰}$  at  $14^\circ\text{S}$ . The  $\delta\text{D}$ -CH<sub>4</sub> reported from 5 latitudinal shipboard sampling transects (between approximately  $55^\circ\text{N}$  and  $65^\circ\text{S}$  in the Pacific Ocean) between March 1989 and October 1995 provided a mean value of  $\delta\text{D}$  of atmospheric CH<sub>4</sub> of  $-86 \pm 3\text{‰}$  ( $-91 \pm 5\text{‰}$  in NH and  $-81 \pm 3\text{‰}$  in SH), with NH data of  $\delta\text{D}$ -CH<sub>4</sub> comparable to our measured values of  $\delta\text{D}$  at NWT and MDO.

[33] Bergamaschi *et al.* [2000] established a 2-year record of CH<sub>4</sub> isotopes at Izaña, Tenerife ( $28^\circ\text{N}$ ,  $16^\circ\text{W}$ , 2370 m) from 1997–1998. Values for  $\delta^{13}\text{C}$ -CH<sub>4</sub> ranged from approximately  $-47.1$  to  $-47.5\text{‰}$  with a peak-to-peak seasonal amplitude of  $0.2\text{‰}$  (from the polyharmonic curve fit) while values of  $\delta\text{D}$  data averaged  $-81\text{‰}$  with a seasonal of  $3.5\text{‰}$ . In a very important finding, they noted that  $\delta\text{D}$  values were in nearly perfect antiphase with mixing ratio

values but that  $\delta^{13}\text{C}$  was out of phase with CH<sub>4</sub> mixing ratios. The apparent difference between mean values of  $\delta\text{D}$  reported by Bergamaschi *et al.* and our work and that of Quay *et al.* [1999] may be the result of problems in comparing data between laboratories with no intercalibration for  $\delta\text{D}$ -CH<sub>4</sub> or may reflect the complexity of local CH<sub>4</sub> sources and sinks affecting a given background air site.

[34] Our measured  $\delta^{13}\text{C}$ -CH<sub>4</sub> values for NWT for the years 1998 and 1999 can be compared directly to values reported by Miller *et al.* [2002] at that site from weekly flask samples. Miller *et al.*'s reported measurements for NWT have a range of values from  $-46.9\text{‰}$  to  $-47.6\text{‰}$  and seasonal amplitudes of  $0.4\text{‰}$  (1998) and  $0.2\text{‰}$  (1999) based on their “smooth curve” fit. Our  $\delta^{13}\text{C}$  average values for 1998–1999 at NWT compare well with that of Miller *et al.* [2002] Monthly mean comparisons of both  $\delta^{13}\text{C}$  and mixing ratio were also in agreement. However, note that the detail (with two maxima per year) in the polyharmonic fits for our two sites, NWT and MDO, agree more closely with the Tenerife  $\delta^{13}\text{C}$  curve fit of Bergamaschi *et al.* [2000] than they do with any of the NH sites reported by Miller *et al.*

[35] Finally, Morimoto *et al.* [2006] have reported measurements of  $\delta^{13}\text{C}$ -CH<sub>4</sub> in air samples from Ny Ålesund, Svalbard ( $79^\circ\text{N}$ ,  $12^\circ\text{E}$ ), from 1996 to 2004. Their scale is  $0.37 \pm 0.04\text{‰}$   $^{13}\text{C}$ -enriched for atmospheric CH<sub>4</sub> samples based on an intercomparison with NIWA. Peak-to-peak amplitude of  $\delta^{13}\text{C}$ -CH<sub>4</sub> was  $0.42\text{‰}$  for 1996–2003, somewhat smaller than the  $0.65\text{‰}$  amplitude observed by Miller *et al.* [2002] at Point Barrow, Alaska ( $71^\circ\text{N}$ ,  $157^\circ\text{W}$ ) for 1998–1999 but comparable to amplitudes at NWT reported in our work.

[36] Our mixing ratio data for CO can be compared to more extensive data sets from the NOAA/GMD network [Novelli *et al.*, 1998a, 1998b, 2003]. Novelli and coworkers established that extratropical Northern Hemispheric sites in the NOAA sampling network exhibit large peak-to-peak amplitudes (as much as 100 ppb at arctic stations [Novelli *et al.*, 1998a]), with seasonal cycles similar to our sites, and significant interannual variability within a site. NOAA data from NWT ranged from 150 to 165 ppb in April and from 120 to 135 ppb in September–October for 1994 [Novelli *et al.*, 1998b] which compares well with our measured values for later years.

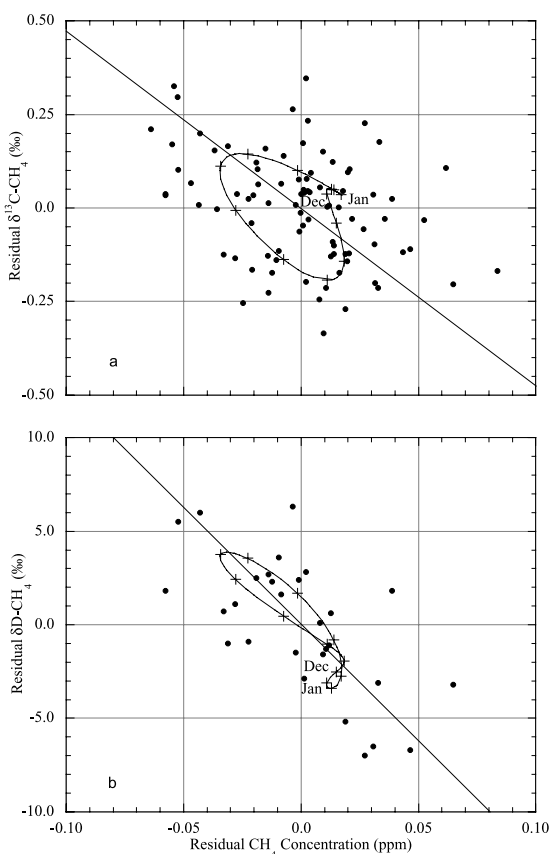
## 5. Discussion

### 5.1. Methane Phase Ellipses and Kinetic Isotope Effects

[37] Bergamaschi *et al.* [2000] and Allan *et al.* [2001a] both described a method to examine seasonal behavior in CH<sub>4</sub> isotope ratios. In order to differentiate between sources and sinks, a “phase ellipse” may be constructed to relate seasonal changes in  $\delta^{13}\text{C}$  to mixing ratio. Assuming a purely aseasonal source function, they derive the following relationship:

$$\Delta\delta \sim \varepsilon(1 + \delta_0)\Delta C/C_0 \quad (2)$$

where  $\varepsilon$  is the kinetic isotope effect ( $k_{13}/k_{12} - 1$ ) and  $\Delta\delta$  and  $\Delta C$  are changes in isotope ratio and mixing ratio respectively about mean values  $\delta_0$  and  $C_0$ . A plot correlating changes in isotope ratio relative to changes in mixing ratio



**Figure 6.** Phase ellipses for CH<sub>4</sub> at Montaña de Oro, California, made by plotting (a) residual  $\delta^{13}\text{C}-\text{CH}_4$  versus residual CH<sub>4</sub> mixing ratio and (b) residual  $\delta\text{D}-\text{CH}_4$  versus residual CH<sub>4</sub> mixing ratio. The slopes of the lines result in kinetic isotope effects for atmospheric loss processes of  $1.010 \pm 0.003$  and  $1.29 \pm 0.07$  for  $k_{12}/k_{13}$  and  $k_{\text{H}}/k_{\text{D}}$ , respectively.

will result in a line, the slope of which is determined by the magnitude of the combined KIEs of CH<sub>4</sub> sink reactions (i.e., “KIE line”). *Allan et al.* [2001a] used a 3D chemical transport model to evaluate the utility of phase ellipses in depicting KIEs and noted, in particular, that off axis distortions were caused by seasonally varying sources resulting in elliptical behavior. Increasing source dominance of the seasonality thus increases the eccentricity of the ellipse.

[38] Figures 5 and 6 show residual  $\delta^{13}\text{C}$  (Figures 5a and 6a) and  $\delta\text{D}$  (Figures 5b and 6b) plotted versus residual mixing ratio for NWT and MDO, respectively. Empirically, the slope of the “KIE line” determined from observational data may be compared with laboratory measured KIEs in the reaction of CH<sub>4</sub> with tropospheric sink processes, i.e., reaction with OH, soil bacteria, and Cl. Fitted curves are derived from seasonal fits, as described earlier in section 4.2, and mid monthly interpolated values are indicated. The linear fits shown, determined from the principle components of the correlation (i.e., the eigenvectors of the covariance matrix), provide the corresponding KIE lines. Slopes of resulting lines at NWT result in KIEs for combined atmospheric loss processes of  $k_{12}/k_{13} = 1.009 \pm 0.003$  and

$k_{\text{H}}/k_{\text{D}} = 1.30 \pm 0.03$  for carbon and hydrogen respectively. Similarly, the slopes of similar lines result in KIEs for atmospheric loss processes of  $k_{12}/k_{13} = 1.010 \pm 0.003$  and  $k_{\text{H}}/k_{\text{D}} = 1.29 \pm 0.07$  at MDO.

[39] The calculated carbon atmospheric KIE at each site is larger than would be expected from a pure OH driven loss process based on laboratory measurements of the carbon KIE for atmospheric CH<sub>4</sub> reaction with OH. The room temperature  $k_{12}/k_{13}$  value has been measured as either 1.0054 [*Cantrell et al.*, 1990] or 1.0039 [*Saueressig et al.*, 2001]. However, the carbon KIEs observed at our sites are consistent with observations by *Allan et al.* [2001b] in the extratropical SH (approximately 1.013 based on data at Scott Base, Antarctica and Baring Head New Zealand) and *Bergamaschi et al.* [2000] in the midlatitude NH (approximately 1.009 at Izaña, Tenerife). Although soil uptake has a higher carbon KIE (1.022 [*Tyler et al.*, 1994]), the estimated upper limit for its global contribution to the CH<sub>4</sub> budget is only approximately 5% of the total sink [e.g., *Gupta et al.*, 1996]. Instead, marine boundary layer Cl oxidation of CH<sub>4</sub> (carbon KIE approximately 1.06) has been identified as a potential explanation for the larger-than-expected carbon KIE observed in  $\delta^{13}\text{C}$  compared to that from primarily OH loss [*Gupta et al.*, 1996; *Allan et al.*, 2001b].

[40] For perspective, at 298 K, the value of the ratio of reaction rate coefficients,  $k_{\text{Cl}}/k_{\text{OH}}$ , for Cl and OH with CH<sub>4</sub> is 15.9, while the ratio of KIE enrichment of CH<sub>4</sub> oxidation by Cl and OH at 298 K is either 12.2 (using 1.0054 for OH KIE) or 16.9 (using 1.0039). Thus even a small amount of marine boundary layer Cl may be an important contribution to the overall carbon KIE of atmospheric CH<sub>4</sub> in spite of its much lesser abundance compared to OH. Our results from NWT and MDO are consistent with a small but significant amount of Cl oxidation of CH<sub>4</sub> in the marine boundary layer. Note that the magnitude of the Cl contribution to the total KIE will impact the slope of the KIE line and add a small off-axis distortion to the isotope phase ellipse [*Allan et al.*, 2001b].

[41] Seasonal variations in stratospheric-tropospheric exchange are also of potential influence to the observed carbon atmospheric KIE. Because oxidation of CH<sub>4</sub> by OH, Cl, and O(<sup>1</sup>D) in the stratosphere enriches CH<sub>4</sub> in <sup>13</sup>C, return of stratospheric air to the troposphere enriches tropospheric  $\delta^{13}\text{C}-\text{CH}_4$  by as much as 1‰ [*Gupta et al.*, 1996; *McCarthy et al.*, 2001; *Wang et al.*, 2002]. Seasonal fluctuations in stratospheric-tropospheric exchange, for example after the springtime breakup of the polar vortex [e.g., *Rex et al.*, 1999], may enrich observed tropospheric CH<sub>4</sub> carbon isotopes annually, providing an inflection in residual  $\delta^{13}\text{C}-\text{CH}_4$  in Figures 3b, 4b, 5a, and 6a.

[42] Atmospheric hydrogen KIEs and their uncertainties calculated from our measurements of mixing ratio and  $\delta\text{D}$  are in reasonable agreement with laboratory determinations of the KIE in the reaction of CH<sub>3</sub>D with OH. Measurements of the hydrogen KIE in the reaction of atmospheric CH<sub>4</sub> with OH put the  $k_{\text{H}}/k_{\text{D}}$  value near 298°K at 1.25 [*Gierczak et al.*, 1997] or 1.294 [*Saueressig et al.*, 2001]. Values at our two sites are also in agreement (within reported uncertainties) with an atmospheric hydrogen KIE reported by *Bergamaschi et al.* [2000] of  $1.23 \pm 0.04$  from samples collected at Izaña, Tenerife. Here again the Cl hydrogen

KIE is larger than the OH one, albeit only a factor of approximately 2 in ratios of KIE enrichments for Cl and OH (i.e., measurements of the hydrogen KIE place the room temperature value at approximately 1.49 [Saueressig et al., 1996; Tyler et al., 2000; Boone et al., 2001]). Therefore the relative impact of Cl on atmospheric  $\delta\text{D-CH}_4$  is lessened somewhat in comparison to analogous reactions of carbon isotopes discussed above, in spite of the aforementioned large  $k_{\text{Cl}}/k_{\text{OH}}$  ratio. Finally, although the carbon KIE in the reaction of CH<sub>4</sub> with soil bacteria is larger than that with OH, the hydrogen KIE is in fact smaller at approximately 1.09 [Snover and Quay, 2000]. Soil uptake will therefore act to lower the overall observed atmospheric CH<sub>4</sub> hydrogen KIE from that of pure OH oxidation, thereby acting in opposition to an increase in the hydrogen KIE from marine boundary layer Cl oxidation of CH<sub>4</sub>. The flux of stratospheric air to the troposphere also acts to enrich tropospheric  $\delta\text{D-CH}_4$  [e.g., Rice et al., 2003], and a seasonal dependence in stratosphere-troposphere exchange, similar to that of  $\delta^{13}\text{C}$ , will likewise affect observed atmospheric CH<sub>4</sub> hydrogen KIEs.

[43] Several additional features of the phase ellipses in Figures 5 and 6 are illustrative. First note the relatively poor correlation between residual  $\delta^{13}\text{C}$  and mixing ratio at both NWT ( $r^2 = 0.10$ ) and MDO ( $r^2 = 0.11$ ). The eccentricities of correlations are 0.925 (ratio of major to minor axis 2.63) at NWT and 0.871 (ratio of major to minor axis 1.930) at MDO. Together these indicate the strong source-driven component to the seasonal cycle in  $\delta^{13}\text{C}$  as a result of the relatively small atmospheric KIE when compared with the  $\delta^{13}\text{C}$  range of atmospheric CH<sub>4</sub> sources. Also of note are the interpolated mid month values that indicate the intra-annual relationship between  $\delta^{13}\text{C}$  and mixing ratio. In particular, the periods January through April (at MDO) and December through March (at NWT) show little variation in either variable, indicating a time when mixing ratio and  $\delta^{13}\text{C}$  appear to be in “pseudosteady state.” Throughout the rest of the year, temporal changes are accompanied by relatively large variations in mixing ratio,  $\delta^{13}\text{C}$ , or in both variables. We made calculations to compare the correlation between the  $\delta^{13}\text{C}$  and mixing ratio values as  $\delta^{13}\text{C}$  was shifted in phase by 1 day intervals. Correlations between the fitted  $\delta^{13}\text{C}$  values and mixing ratio values were best when  $\delta^{13}\text{C}$  was shifted in phase by approximately 3 months (89 days at MDO with  $r^2 = 0.91$  and 101 days at NWT with  $r^2 = 0.72$ ) rather than the 6-month phase shift expected in a sink only model (see below).

[44] In contrast, seasonal changes in  $\delta\text{D}$  appear better correlated with seasonal changes in mixing ratio (correlation coefficients,  $r^2$ , are 0.72 at NWT and 0.42 at MDO). The large atmospheric hydrogen KIE acting on  $\delta\text{D-CH}_4$  is thought to be the basis of this well-correlated relationship [e.g., Bergamaschi et al., 2000]. The tightness of the data about the major axis of the ellipse results in eccentricities of 0.991 (ratio of major to minor axis 7.51) and 0.999 (ratio of major to minor axis 22.6) at NWT and MDO, respectively. Nonetheless, seasonal behavior in  $\delta\text{D}$  is not absent of source effects (see our discussion of phases in the next section) and there are several off axis features at each site.

[45] Finally, we note considerable differences with  $\delta^{13}\text{C}$  phase ellipses drawn from data in the extratropical SH [Allan et al., 2001a]. Results from analyses at Baring Head,

New Zealand show considerably tighter correlations between  $\delta^{13}\text{C}$  and mixing ratio and stronger sink influence on  $\delta^{13}\text{C}$  values. Northern hemispheric results of Bergamaschi et al. [2000] from Izaña, Tenerife, display very similar behavior in both  $\delta^{13}\text{C}$  and  $\delta\text{D}$  to results presented here, i.e., poor correlations between  $\delta^{13}\text{C}$  and mixing ratio and strong correlations between  $\delta\text{D}$  and mixing ratio. The difference in  $\delta^{13}\text{C}$  behavior between the NH and SH can largely be attributed to an isolation of the SH from seasonal variability in northern hemispheric sources which are attenuated through interhemispheric transport.

## 5.2. A Simple Sink-Only Model for Methane Seasonality

[46] Following the conceptual model of Bergamaschi et al. [2000] and Allan et al. [2001a], we constructed a simple time-dependent 1-box model to calculate the seasonal effects on northern hemispheric CH<sub>4</sub> from OH alone (i.e., the model assumes that sources are unchanging and are input to maintain steady state over a year), similar to equation (2) above. Observed deviations from model predictions can largely be attributed to seasonally variant sources in the NH and transport associated with these sources. If we assume a strictly aseasonal source function, then variations in CH<sub>4</sub> about its average value  $[\text{CH}_4]_0$  can be described by the seasonally cycling OH mixing ratio independent of sources and transport. For small perturbations in CH<sub>4</sub>,  $\delta[\text{CH}_4]/dt = k[\text{CH}_4]_0[\text{OH}]'$  where  $[\text{OH}]'$  and  $[\text{CH}_4]'$  are perturbations in OH and CH<sub>4</sub> about their mean values. Differential equations were solved for  $[\text{CH}_4]'$ ,  $\delta^{13}\text{C-CH}_4$ , and  $\delta\text{D-CH}_4$ , where the lifetime of CH<sub>4</sub> is prescribed to be 8 years, perturbations in OH mixing ratio were derived from output of the OSLO 3D CTM at 36°N (I. Isaksen, University of Oslo, Norway, personal communication, 2001), and KIEs were taken from Cantrell et al. [1990] and Gierczak et al. [1997] at 277 K.

[47] This sink-only model predicts CH<sub>4</sub> mixing and both isotope ratios to be anticorrelated, with the mixing ratio minimum and corresponding isotope ratio maximum in October. The maximum value in mixing ratio and corresponding minimum in either isotope ratio appear in April. The observed phase lag between maximum OH concentrations and minimum CH<sub>4</sub> mixing ratios of 3 months (i.e.,  $\pi/2$ ) is inherent in long-lived chemical species responding to a sinusoidal forcing [Camp et al., 2001]. Despite the simplicity of our model, it is validated, in part, because of its similarity in results to mixing and isotope ratio models of considerably higher order in both phase and amplitude of seasonal cycles [e.g., Gupta et al., 1996; Allan et al., 2001a]. Allan et al. [2001b] pointed out the importance of marine boundary layer Cl to the seasonal cycle in  $\delta^{13}\text{C}$  in the extratropical SH. We note that although including Cl increases the amplitude by more than a factor of two, the phase of the  $\delta^{13}\text{C}$  cycle is changed by 1 month or less [see Allan et al., 2001b, Figure 3].

## 5.3. Inferences Drawn From Observations and Our Sink-Only Model Results

[48] The significant disagreement between measured CH<sub>4</sub> seasonal cycles (Figures 3 and 4) and modeled CH<sub>4</sub> sink-based seasonal cycles can be attributed to the seasonal behavior in sources of atmospheric CH<sub>4</sub>. Observed season-



**Table 1.** Approximate Annual Global Emission Strengths and Corresponding  $\delta^{13}\text{C}$  and  $\delta\text{D}$  Values for Various Categories of CH<sub>4</sub> Sources Circa 1998<sup>a</sup>

| Source                        | Emission Strength,<br>Tg/yr (10 <sup>12</sup> g/yr) | $\delta^{13}\text{C}$ , <sup>b</sup><br>‰ | $\delta\text{D}$ , <sup>c</sup><br>‰ |
|-------------------------------|-----------------------------------------------------|-------------------------------------------|--------------------------------------|
| Tundra                        | 10                                                  | −65 <sup>d</sup>                          |                                      |
| Lakes (bogs, fens, and lakes) | 75                                                  | −59 <sup>d</sup>                          | −322                                 |
| Swamps and marshes            | 106                                                 | −56                                       | −322                                 |
| Rice paddies                  | 80                                                  | −59                                       | −323                                 |
| Animals                       | 95                                                  | −62                                       | −305                                 |
| Landfills                     | 42                                                  | −50                                       | −293                                 |
| Gas venting and leakage       | 54                                                  | −38                                       | −175                                 |
| Oceans (coastal margins)      | 10                                                  | −40                                       |                                      |
| Coal minings                  | 27                                                  | −37                                       | −175                                 |
| Biomass burning               | 45                                                  | −26                                       | −210                                 |
| Termites                      | 20                                                  | −57                                       |                                      |
| Gas hydrates                  | 10                                                  | −52                                       |                                      |
| Total                         | 574                                                 |                                           |                                      |

<sup>a</sup>See Gupta *et al.* [1996, and references therein], Snover *et al.* [2000, and references therein], and Ehhalt *et al.* [2001, and references therein].

<sup>b</sup>Weighted average  $\delta^{13}\text{C}$  of these sources is approximately −52‰.

<sup>c</sup>Weighted average  $\delta\text{D}$  of these sources is approximately −284‰.

<sup>d</sup>Between 60°N and 90°N  $\delta^{13}\text{C}$  values are decreased to −70‰.

ality is both larger in amplitude and out of phase with that predicted based a purely OH sink-driven seasonality. At both NWT and MDO, mixing ratio has a relatively flat maximum occurring late fall through early spring contrasting to our sink-only model which predicts a more distinct maximum near April. Springtime draw down of mixing ratio occurs roughly as anticipated, but is shorter and has an abrupt minimum in late summer; the model predicts minimum values near October. Differences in  $\delta^{13}\text{C}$  are even more striking. The most highly  $^{13}\text{C}$ -enriched values of CH<sub>4</sub> occur during summer at MDO and NWT, a full quarter phase before the model predicts the most  $^{13}\text{C}$ -enriched CH<sub>4</sub> to occur. Perhaps even more compelling evidence of a source-driven cycle is that the most  $^{13}\text{C}$ -depleted CH<sub>4</sub> occurs during October, when the most  $^{13}\text{C}$ -enriched values are expected based on the model. Although clearly the seasonally variant OH sink influences observed  $\delta^{13}\text{C}$ , the impact appears to be small because of the relatively small carbon KIE. As noted above,  $\delta\text{D}$ -CH<sub>4</sub> exhibits a clear anticorrelation with CH<sub>4</sub> mixing ratio and has a stronger sink-driven component than does  $\delta^{13}\text{C}$ . As with mixing ratio, however, the cycle in  $\delta\text{D}$  is clearly not purely sink-driven. The most D-enriched CH<sub>4</sub> occurs during midsummer and values most depleted occur in midwinter; model predictions of  $\delta\text{D}$  are a full quarter phase later than our observed  $\delta\text{D}$  values.

[49] Table 1 lists the major CH<sub>4</sub> sources and their approximate annual strengths,  $\delta^{13}\text{C}$ , and  $\delta\text{D}$  signatures. The two CH<sub>4</sub> sources most likely to influence midlatitude NH background air in such a way as to drive a seasonal cycle are wetlands and biomass burning. To provide background for the following discussion of our data, we describe general features of each of these sources in turn.

[50] Methane emissions in both natural wetlands [e.g., Matthews, 2000, and references therein] and agricultural wetlands such as rice paddies [Neue, 1992] are likely to exhibit seasonality. In all wetlands, anaerobic bacteria operate to produce CH<sub>4</sub>, either from fermentation of labile organic substrates or from reduction of CO<sub>2</sub> with H<sub>2</sub>. Warmer weather and hence warmer soil temperatures play

a part as does the availability of CH<sub>4</sub> precursors. Furthermore seasonal flooding, either human induced as in the case of some rice paddies, or naturally occurring, as in some rice paddies and many wetland areas, can change the areal extent of the anaerobic domain.

[51] Wetland rice from paddy fields is generally planted in late spring with a harvest in late summer or early fall although there are exceptions in areas of the world where double cropping takes place [e.g., Neue, 1992; Sass, 1994; Minami, 1995, and references therein]. In many cases, late season CH<sub>4</sub> emissions in rice paddies provide the maxima or one of a few relative maxima in emissions over the whole growing season [e.g., Schütz *et al.*, 1989]. While there can be much variability both interannually and intra-annually in CH<sub>4</sub> emissions and their isotope ratios during the growing season (depending on factors such as water management, organic and inorganic fertilizers applied, region of the world, etc.), isotopically depleted CH<sub>4</sub> is preferentially emitted for both  $^{13}\text{C}$  and D isotopes in comparison to the atmospheric value [e.g., Tyler *et al.*, 1997]. An emissions maximum from rice agriculture in late summer is compatible with high atmospheric CH<sub>4</sub> mixing ratios and low  $\delta^{13}\text{C}$  and  $\delta\text{D}$  values observed in early fall and contrasts expected CH<sub>4</sub> mixing and isotope ratio behavior based on a sink-only model.

[52] In natural wetlands such as lakes, bogs, fens, and swamps, emissions follow a seasonal cycle which is in part temperature-dependent [e.g., Whalen and Reeburgh, 1988; Bellisario *et al.*, 1999] and water-table-dependent [e.g., Devol *et al.*, 1988; Bartlett *et al.*, 1992; Roulet *et al.*, 1993; Strack *et al.*, 2004], hence seasonally dependent. Here, too, a range of isotopic values for CH<sub>4</sub> have been measured for emissions, but in virtually all cases,  $^{13}\text{C}$ -depleted CH<sub>4</sub> is emitted for both C and H isotopes in comparison to the atmospheric value [e.g., Quay *et al.*, 1988; Tyler *et al.*, 1988; Tyler, 1991; Lansdown, 1992; Martens *et al.*, 1992]. High northern latitude sources such as seasonally frozen tundra soils and bog and fen peatlands tend to begin releasing CH<sub>4</sub> in late spring and reach their maximum CH<sub>4</sub> emissions during the late summer or early fall [Whalen and Reeburgh, 1992]. High latitude natural wetland emissions are compatible with high CH<sub>4</sub> mixing ratios and low values of  $\delta^{13}\text{C}$  and  $\delta\text{D}$  observed in early fall. In contrast to high-latitude wetlands, tropical wetlands such as swamps and floodplains often emit CH<sub>4</sub> at all times of the year. However, the magnitude of emissions may still be strongly seasonal, with areal extent of emissions depending on the water table [e.g., Devol *et al.*, 1988; Keller, 1990; Matthews, 2000], particularly in floodplain regions in response to precipitation cycles.

[53] Seasonal patterns in biomass burning and their effect on CH<sub>4</sub> emissions, are more readily seen by first studying burning and its relationship to CO mixing ratio. This is because biomass burning is a major source of CO emissions, especially in the SH where fewer anthropogenic sources of CO exist. Overall, it makes up some 20–25% of the annual CO source budget globally [Seiler and Conrad, 1987; Khalil and Rasmussen, 1990; Suntharalingam *et al.*, 2005]. Its effect on CH<sub>4</sub> mixing ratio will not be as strong because as Table 1 shows, biomass burning is only about 10% of the overall source budget. Biomass burning includes



both annual land clearing for agriculture, typically in spring and natural fires.

[54] In order to understand the seasonal pattern of biomass burning on CO mixing ratio, one must consider the relationship between CO mixing ratio and the two dominant varying effects on it, the OH sink and biomass burning, together. Generally, the maximum seasonal CO levels occur in late winter or early spring with a lag of several months after the seasonal OH minimum [Novelli *et al.*, 1998a]. Likewise, lowest CO values occur in late summer because of maximum OH concentrations. Other important factors for CO in the NH which will impact our two Northern Hemispheric CH<sub>4</sub> measurement sites are the transport of atmospheric pollutants from northern Europe and Siberia into the polar vortex where they accumulate in winter and early spring [Rahn and Lowenthal, 1986] and seasonal emissions of biomass burning in the northern tropics, which typically reach their maximum levels in February and March.

[55] Given the above background on biomass burning and its relationship to CO data, we can examine the effect of its seasonality on CH<sub>4</sub> mixing and isotope ratios. The timing of northern tropical season emissions from biomass burning, typically a maximum in February and/or March [Novelli *et al.*, 1998a], is consistent with the relative maxima in CH<sub>4</sub> that we see at both NWT (approximately April) and MDO (approximately March) in Figures 3a and 4a. Each of these signals then begins to drop quickly beginning in mid spring as OH values rise toward their summer maxima. As shown in Table 1, the combination of  $\delta^{13}\text{C}$  and  $\delta\text{D}$  signatures of CH<sub>4</sub> from biomass burning are among the very few which are significantly different compared to the majority of other CH<sub>4</sub> sources. We should expect to see some shift toward more <sup>13</sup>C-enriched CH<sub>4</sub> at the same time in spring as the mixing ratio increases. In fact this shift is seen at NWT in April (Figure 3b) but it continues toward its maximum value in June with the advent of rising OH mixing ratios as summer approaches. At MDO a similar rise in  $\delta^{13}\text{C}$  begins in March (Figure 4b) which again continues toward its June maximum.

[56] In contrast to the  $\delta^{13}\text{C}$  signal, the  $\delta\text{D}$  signal from biomass burning is not very different from the  $\delta\text{D}$  values from natural gas and coal sources of CH<sub>4</sub>, the two fossil fuel sources. Monthly average  $\delta\text{D}$  values shown in Figures 3c and 4c indicate an inflection point at NWT in March but nothing out of the ordinary at MDO. In both cases,  $\delta\text{D}$  values rise rapidly toward their summer maxima after early spring. We surmise that the biomass burn  $\delta\text{D}$  signal is not separable from  $\delta\text{D}$  values of some of the fossil fuel sources. In fact, the two types of sources may have a canceling effect on each others' seasonal  $\delta\text{D}$  impact at certain times of the year. For instance, a proposed shift in  $\delta\text{D}$  of CH<sub>4</sub> from increased biomass burning as spring comes could also be from decreasing fossil fuel use for heating as warmer weather ensues.

[57] To summarize our description of the effect of seasonal sources on the seasonal behavior of our CH<sub>4</sub> and CO mixing ratio data, the mixing and isotopic ratio evidence indicates seasonally variant isotopically depleted CH<sub>4</sub> sources peaking in late summer or early fall, likely from natural or cultivated wetland emissions. Although the intensity of fires and controlled biomass burning vary greatly from year to year [e.g., Novelli *et al.*, 2003], they

are an annual event with a seasonal pattern notwithstanding extraordinary fire events. Our results point to the presence of isotopically enriched sources peaking in late spring possibly from biomass burning during annual land clearing before planting.

[58] As a final caveat, we note that to achieve a more rigorous exploration of seasonal wetland and biomass burning CH<sub>4</sub> sources than is presented here, as well as the multiple chemical sinks of atmospheric CH<sub>4</sub>, a multidimensional atmospheric chemical transport model including dual isotope tracers would be useful. Time-dependent advanced model calculations coupled to our observational data are the best method to explore the time lag for the mix of atmospheric CH<sub>4</sub> and CO sources reaching our northern hemispheric sites as well as for exploring the observed offsets in the correlated and anticorrelated maximum-minimum seasonalities, mixing ratio,  $\delta^{13}\text{C}$ , and  $\delta\text{D}$  of CH<sub>4</sub>. However, the features exhibited by the seasonal cycles in CH<sub>4</sub> and CO mixing ratio and  $\delta^{13}\text{C}$  and  $\delta\text{D}$  of CH<sub>4</sub> are robust and appear representative of the extra tropical Northern Hemisphere given their similarity to data presented from complementary Northern Hemispheric sites [Quay *et al.*, 1999; Bergamaschi *et al.*, 2000; Morimoto *et al.*, 2006].

## 6. Conclusions

[59] Measurements of stable isotope ratios in atmospheric CH<sub>4</sub> provide a valuable investigative tool to gain a comprehensive understanding of the global budget of CH<sub>4</sub>. Our record of  $\delta^{13}\text{C}$ -CH<sub>4</sub> from the continental United States represented by sites at Niwot Ridge, Colorado, and Montaña de Oro, California, indicates clear and consistent seasonal behavior in  $\delta^{13}\text{C}$ -CH<sub>4</sub> values from 1995 to 2001. However, there was relatively poor correlation of  $\delta^{13}\text{C}$  with seasonal behavior in CH<sub>4</sub> mixing ratio over this period. We attribute this to the relatively small carbon KIE in the reaction with OH, providing a  $\delta^{13}\text{C}$ -CH<sub>4</sub> signal that is in large part source driven in the NH.

[60] A more recently initiated time series of measurements of  $\delta\text{D}$ -CH<sub>4</sub> were begun during 1998 at Niwot Ridge and early in 2000 at Montaña de Oro, extending through 2001 at each site. In contrast to the results of the  $\delta^{13}\text{C}$  data, seasonal behavior in  $\delta\text{D}$ -CH<sub>4</sub> displayed a strong anticorrelation with seasonal variations in CH<sub>4</sub> mixing ratio, suggesting that  $\delta\text{D}$  is a stronger indicator of sink effects. This is because of the large hydrogen KIE in CH<sub>4</sub> sink reaction. While it is clear that seasonal behavior in both isotopic variables are influenced by seasonal cycles in both sources and sinks, results presented here demonstrate  $\delta^{13}\text{C}$  to be a better indicator of source-driven seasonality and  $\delta\text{D}$  to be a better indicator of sink-driven seasonality.

[61] In considering the sink-driven isotopic seasonal cycles, similarities in the average annual CO and CH<sub>4</sub> mixing ratios, and CH<sub>4</sub> isotope ratios for the two most important seasonally varying CH<sub>4</sub> sources (biomass burning and wetland emissions), our atmospheric CH<sub>4</sub> data strongly indicate a seasonal cycle in CH<sub>4</sub> sources that is weighted toward enhanced springtime CH<sub>4</sub> emissions (likely from a <sup>13</sup>C-enriched biomass burning source such as land clearing for spring planting) and additional late summer source enhancement with <sup>13</sup>C-depleted values in the late summer

and early fall (possibly from natural or anthropogenic wetland sources peaking at this time).

[62] Having both C and H isotopic information can help uncover the seasonal behavior of atmospheric CH<sub>4</sub> sources by decoupling the seasonality of CH<sub>4</sub> sink processes. Continued measurements of both stable isotopes as well as CH<sub>4</sub> mixing ratio at multiple locations coupled with advanced calculations from atmospheric models will ultimately provide the best means to quantify seasonal cycles and the strengths of CH<sub>4</sub> sources and to understand their global distributions. Future intercomparisons between multiple research groups making similar isotope measurements worldwide would be particularly useful for data assimilation and modeling.

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